

## Machine Learning Assignment

### Question 3: Breast Cancer Classification - Confusion Matrix Analysis

#### Given Data:

Total samples = 1800 (malignant) + 1800 (benign) = 3600 samples. Test set size = 20% of 3600 = 720 samples. True Positives (TP) = 210, True Negatives (TN) = 190, False Positives (FP) = 30.

#### Step 1: Finding the missing False Negatives (FN)

Since the confusion matrix must account for all samples in the test set, the sum of TP, TN, FP, and FN must equal the total number of test samples (720).

$$FN = \text{Total Test Samples} - (TP + TN + FP)$$

$$FN = 720 - (210 + 190 + 30) = 720 - 430 = 290$$

**False Negatives (FN) = 290**

#### Step 2: Confusion Matrix

|                   | Predicted: Malignant | Predicted: Benign |
|-------------------|----------------------|-------------------|
| Actual: Malignant | TP = 210             | FN = 290          |
| Actual: Benign    | FP = 30              | TN = 190          |

#### Step 3: Performance Metrics

$$\text{Accuracy} = (TP + TN) / \text{Total Samples}$$

$$= (210 + 190) / 720 = 400 / 720 = 0.5556 \rightarrow \mathbf{55.56\%}$$

$$\text{Precision} = TP / (TP + FP)$$

$$= 210 / (210 + 30) = 210 / 240 = 0.875 \rightarrow \mathbf{87.5\%}$$

$$\text{Sensitivity (Recall)} = TP / (TP + FN)$$

$$= 210 / (210 + 290) = 210 / 500 = 0.42 \rightarrow \mathbf{42\%}$$

$$\text{Specificity} = TN / (TN + FP)$$

$$= 190 / (190 + 30) = 190 / 220 = 0.8636 \rightarrow \mathbf{86.36\%}$$

| Metric               | Value  |
|----------------------|--------|
| Accuracy             | 55.56% |
| Precision            | 87.5%  |
| Sensitivity (Recall) | 42%    |
| Specificity          | 86.36% |

## Interpretation / Effectiveness in Medical Diagnosis:

The classifier shows high Precision (87.5%) and high Specificity (86.36%), meaning it is quite reliable when it predicts a case as malignant and it correctly identifies most benign (healthy) cases, keeping false alarms low. However, the Sensitivity (Recall) is only 42%, which is a serious concern in a medical diagnosis context. A low sensitivity means the model misses more than half of the actual malignant (cancer) cases, classifying them as benign (290 False Negatives). In cancer screening, a False Negative is far more dangerous than a False Positive because a missed cancer diagnosis can delay life-saving treatment. Therefore, despite a reasonable overall accuracy, this model is not clinically dependable in its current form and would need to be improved (e.g., by adjusting the classification threshold, handling class imbalance, or using better feature representations) to increase sensitivity before being used for real-world diagnosis.

## Question 5: Training vs Validation Accuracy Analysis

| Epoch | Training Accuracy | Validation Accuracy |
|-------|-------------------|---------------------|
| 1     | 60%               | 58%                 |
| 2     | 65%               | 63%                 |
| 3     | 70%               | 68%                 |
| 4     | 75%               | 72%                 |
| 5     | 80%               | 73%                 |
| 6     | 85%               | 72%                 |
| 7     | 88%               | 70%                 |
| 8     | 92%               | 68%                 |

### (i) Epoch at which overfitting starts

Validation accuracy increases steadily along with training accuracy up to Epoch 5 (reaching its peak of 73%). From **Epoch 6 onward**, validation accuracy starts decreasing (72% → 70% → 68%) even though training accuracy keeps rising (85% → 88% → 92%). This growing gap between training and validation accuracy indicates that the model begins **overfitting from Epoch 6** (Epoch 5 being the last epoch where validation performance was still improving / at its peak).

### (ii) Why validation accuracy decreases while training accuracy increases

As training progresses beyond the optimal point, the model starts to memorize the specific patterns, noise, and minor details of the training data rather than learning the general underlying relationships in the data. This causes the training accuracy to keep climbing, since the model fits the training set almost perfectly. However, because these memorized patterns do not generalize to unseen data, the model's performance on the validation set degrades. In other words, the model's capacity is being used to fit training-set-specific noise instead of learning generalizable features, so the gap between training and validation accuracy widens - this is the classic signature of overfitting.

### (iii) Two methods to improve validation performance

- **Early Stopping:** Monitor the validation accuracy/loss during training and stop training once validation performance stops improving (around Epoch 5 in this case), instead of continuing to train until Epoch 8. This prevents the model from over-training on the training data.
- **Regularization Techniques:** Apply methods such as Dropout, L1/L2 weight regularization, or data augmentation to reduce the model's tendency to memorize the training data. Reducing model complexity (fewer layers/parameters) or increasing the size/diversity of the training data can also help the model generalize better to unseen validation and test data.